Rotorcraft Emerge

A worldwide race was under way in the 1930s to perfect the helicopter, but it was only when disparate touches of genius were brought together that progress was made. Spain's Juan de la Cierva invented the hinge systems that made rotors practical; Austrian Raoul Hafner came up with the cyclic system that made them controllable; Frenchman Louis Breguet created coaxial contra-rotating blades that prevented the rotor blades and the helicopter fuselage from rotating in opposite directions (torque reaction); and Russian-American Igor Sikorsky made the important steps that turned the autogyro into a true helicopter.

△ Cierva C19 1930
Origin Spain/UK
Engine 80 hp Armstrong Siddeley Genet II radial
Top speed N/A

A method of spinning the main rotor by deflecting the propeller wash allowed the C19 to "spin up" while stationary. In the MkVI, the rotor was "prespun" directly by the engine.

▽ Cierva C8 MkIV Autogiro 1930
Origin UK
Engine 200 hp Armstrong Siddeley Lynx IVC 7-cylinder radial
Top speed 100 mph (161 km/h)

Cierva's "articulated" rotor is now used on almost all helicopters. The C8 added drag dampers to limit blade movement and successfully completed a 3,000-mile (4,828-km) tour of Britain.

△ Cierva C30A Autogiro 1934
Origin Spain/UK
Engine Armstrong Siddeley Genet Major 1A radial
Top speed 110 mph (177 km/h)

Cierva's autogyros were rightly described in their time as "the most important step in aeronautics since the Wright Brothers." Tragically, he was killed in a plane crash in 1936.

△ D'Ascanio D'AT3 1930
Origin Italy
Engine 95 hp Fiat A-505 piston
Top speed N/A

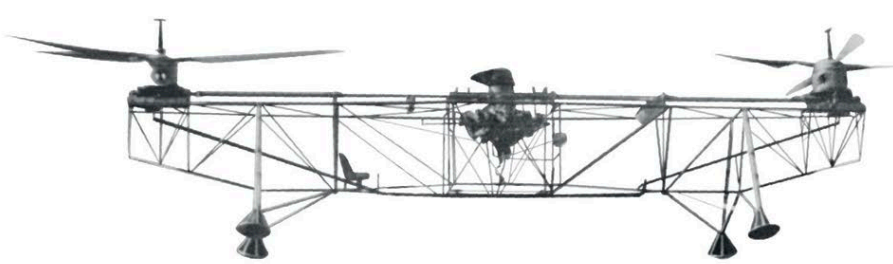
This early coaxial twin rotor machine set height (59 ft/18 m) and distance (3,537 ft/1,078 m) records, but designer Corradino D'Ascanio's potential was unfulfilled. He went on to invent the first scooter.

◁ de Havilland/Cierva C24 Autogiro 1931
Origin UK
Engine 120 hp de Havilland Gipsy III in-line
Top speed 115 mph (185 km/h)

The sole venture of de Havilland into autogyros "married" a Cierva rotor to a DH Puss Moth fuselage. Designed for three people, it could barely lift two, and only one was made.

△ Herrick HV-2A Verta 1933
Origin USA
Engine 125 hp Kinner B-5 radial
Top speed 99 mph (159 km/h)

The Verta was an inspired biplane design with a conventional lower wing and an upper wing that rotated, autogyro style, for low-speed flight or landings. It was too heavy to develop successfully.



△ Florine Tandem Motor 1933
Origin Belgium
Engine 180 hp Hispano-Suiza piston
Top speed N/A

Russian émigré Nicolas Florine built the first flyable twin tandem rotor helicopter. Rather than contra-rotating, the rotors were tilted 10 degrees in relation to each other to counter torque.